CORAL run 2026-10-09_102552 — results (absolute-throughput scoring)

Run summary

Item	Value
Task	cache-ext-get-scan (LevelDB GET/SCAN, cache_ext eviction-policy search)
Scoring in this run	score = measured throughput in ops/s (absolute; higher is better). No baseline calibration, z-score, tanh or weights; branch absolute-throughput-score (commit e31539d).
Run start → stop (local)	10:25:52 → 10:35:17 (9.4 min)
Stop reason	max_real_attempts (cap 5, real attempts 5); triggering attempt 1a7504b5, score 1199.98
Agent	1 agent (captain-nemo), runtime claude_code, model alias claude-opus-5-5 → Bedrock us.anthropic.claude-opus-5-5
Attempts recorded	5 = 5 real + 0 --tune; coral eval invocations: 6; grader errors: 0 (one more eval invocation than recorded attempts)
Attempts by code	4 ran unmodified reference policies (get_scan.c, s3fifo.c, mru.c, sampling.c); 1 agent-evolved design
Attempt status (CORAL)	improved: 2, regressed: 3 (relative to the agent's own previous best)
Agent sessions (claude -p)	5; terminal reason: aborted_streaming 5
LLM usage (agent)	14,927 output tokens; 1,812,949 cache-read + 101,472 cache-write + 54 input tokens; Claude Code cost estimate $1.17; API time 285 s
Notes / skills written	0 notes (2 write attempts were denied by the permission layer); skills present: create-notes, deep-research, organize-files, skill-creator

Headline numbers

Metric	Value	Metric	Value
Best score (= throughput)	#3 b171f393 mru.c (unmodified): 1,258.70 ops/s	Throughput, mean ± sd of 5	1,227.4 ± 21.0 ops/s
Best agent-evolved attempt	#4 0e985cef: 1,230.52 ops/s	Throughput range (max − min)	58.7 ops/s (4.78% of mean)
Worst score	#5 sampling.c (unmodified): 1,199.98 ops/s	Final attempt (#5)	sampling.c (unmodified): 1,199.98 ops/s
Reference: run-1 noop baseline (external)	1,224.3 ± 9.5 ops/s (n=8, run 2026-10-09_004013)	Attempts above that reference mean	3 / 5

This run contains no noop evaluation (the new scoring needs no baseline). Where a baseline is shown it is the 8-run noop calibration from the first run on the same node and workload (z-score scoring, 2026-10-09_004013) and is marked 'external'.

Scoring and recorded probes

Probe	Source	Role in this run	Unit
throughput	results.json: throughput_ops_per_sec	THE SCORE (maximized)	ops/s
rbytes	cgroup io.stat (delta over measured phase)	diagnostic only	MiB
wbytes, rios	cgroup io.stat	diagnostic only	MiB, count
refault_file, pgmajfault, pgactivate, resident_file	cgroup memory.stat	diagnostic only	count, count, count, MiB
wallclock	benchmark wall time (setup + warm-up + measured)	diagnostic only	s

Environment and workload

Evaluation node (CloudLab, Wisconsin)

Component	Value	Component	Value
Node type	c220g1	Kernel	6.6.8-cache-ext+ (Ubuntu 22.04.2)
CPU	2 × Intel Xeon E5-2630 v3 @ 2.40 GHz	Cores / threads	16 cores / 32 threads (2 sockets, 2 NUMA nodes)
Memory	131,732,952 kB (125.6 GiB), no swap	Disks	2 × 1.1 TB SAS HDD, 1 × 447 GB SSD
Benchmark filesystem	/mydata on /dev/sda3 (HDD), 63 GB	Cgroup	cgroup v2; benchmark memory.max = 64 MB
Compiler / tools	clang 14.0.0-1ubuntu1.1, bpftool v7.3.0	Eval isolation	one eval at a time (flock); policy attached via BPF struct_ops

Workload parameters (benchmark harness defaults, every attempt)

Parameter	Value	Parameter	Value
Benchmark	My-YCSB (leveldb-scan branch), run_leveldb	Request distribution / Zipfian constant	zipfian / 0.99
Operation mix	95% GET (read), 5% SCAN	Worker threads	4 (exactly 1 dedicated scan thread)
Scan length	500 keys per SCAN	Measured phase	30 s
Database entries	500,000	Warm-up phase	10 s (same mix, probes not recording)
Key / value size	16 B / 200 B	LevelDB block cache	16 MiB
Cgroup memory limit (page cache)	64 MB	Page cache at start	dropped before each run
Per-eval timeout	180 s	Observed eval wall time	39.8 s mean (39.6 … 39.9)

Per-attempt results

#	Local	Commit	Parent	Policy	Status	SCORE = throughput (ops/s)	Δ ops/s vs ext. ref	Δ%	Wall s	Eval s	+/− lines	Code lines
1	10:26:45	17a0a2bd	d14abdbe	get_scan.c (unmodified)	improved	1,225.18	+0.9	+0.07	39.9	44	+509/−50	603
2	10:29:24	938ee028	17a0a2bd	s3fifo.c (unmodified)	regressed	1,222.54	-1.8	-0.15	39.9	46	+409/−410	602
3	10:30:36	b171f393	938ee028	mru.c (unmodified)	improved	1,258.70	+34.4	+2.81	39.6	46	+139/−438	303
4	10:32:58	0e985cef	b171f393	evolved v1	regressed	1,230.52	+6.2	+0.51	39.7	46	+146/−80	369
5	10:34:29	1a7504b5	0e985cef	sampling.c (unmodified)	regressed	1,199.98	-24.3	-1.99	39.7	46	+224/−129	464

Shaded = highest score. Δ columns compare with the external run-1 noop mean (1,224.3 ops/s). Eval s = submit → graded; with no calibration step every attempt takes one evaluation. Status is CORAL's comparison with the agent's own previous best, not with a baseline.

Diagnostic probes and derived per-operation metrics

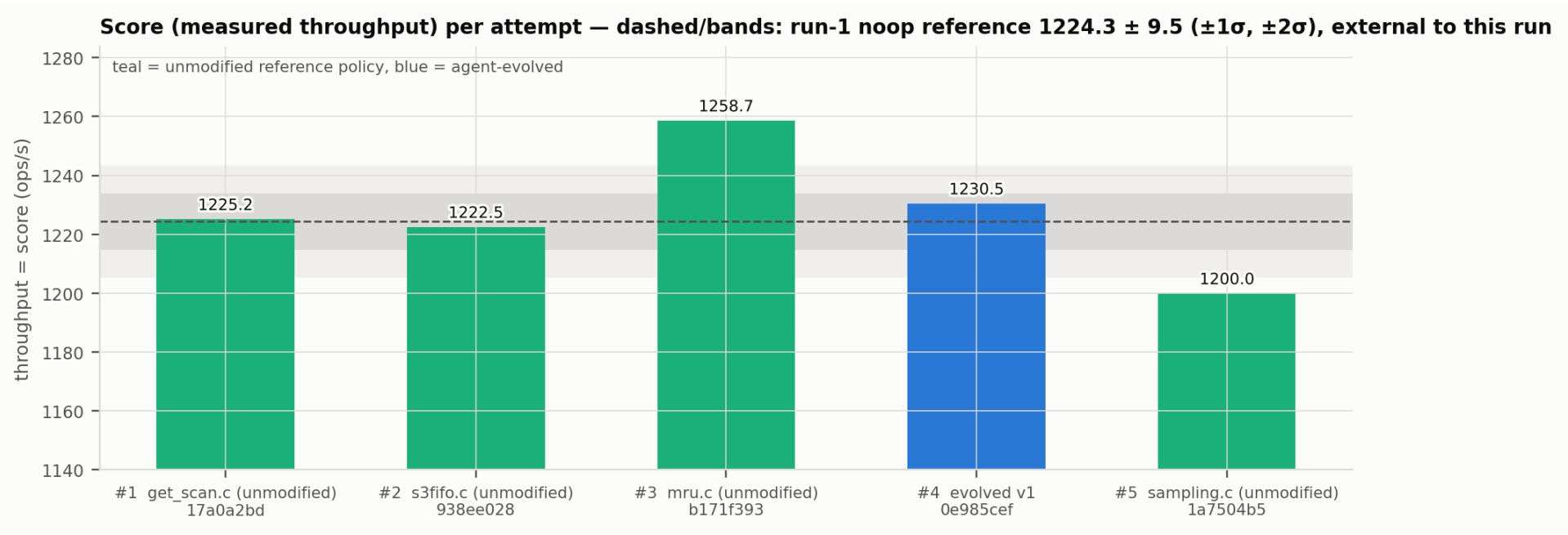
#	Commit	Policy	Bytes read MiB	Δ% vs ext. ref	Bytes written MiB	Read I/Os	Refaults	Δ% vs ext. ref	pgmajfault	pgactivate	Resident file MiB	Ops in phase (≈ tput × 30)	Read KB / op	Refaults / 1k ops	Read I/Os / 1k ops	Read MB/s
1	17a0a2bd	get_scan.c (unmodified)	302.7	-0.5	0.0	22,154	50,745	-0.6	31	14,071	47	36,755	8.64	1,380.6	602.7	10.58
2	938ee028	s3fifo.c (unmodified)	314.3	+3.4	0.0	22,183	53,676	+5.1	31	14,153	47	36,676	8.99	1,463.5	604.8	10.99
3	b171f393	mru.c (unmodified)	300.7	-1.1	0.0	24,461	50,381	-1.3	16	13,573	45	37,761	8.35	1,334.2	647.8	10.51
4	0e985cef	evolved v1	291.1	-4.3	0.0	23,718	47,889	-6.2	16	12,234	45	36,916	8.27	1,297.3	642.5	10.17
5	1a7504b5	sampling.c (unmodified)	281.5	-7.4	0.0	23,075	45,427	-11.0	31	11,125	48	35,999	8.20	1,261.9	641.0	9.84

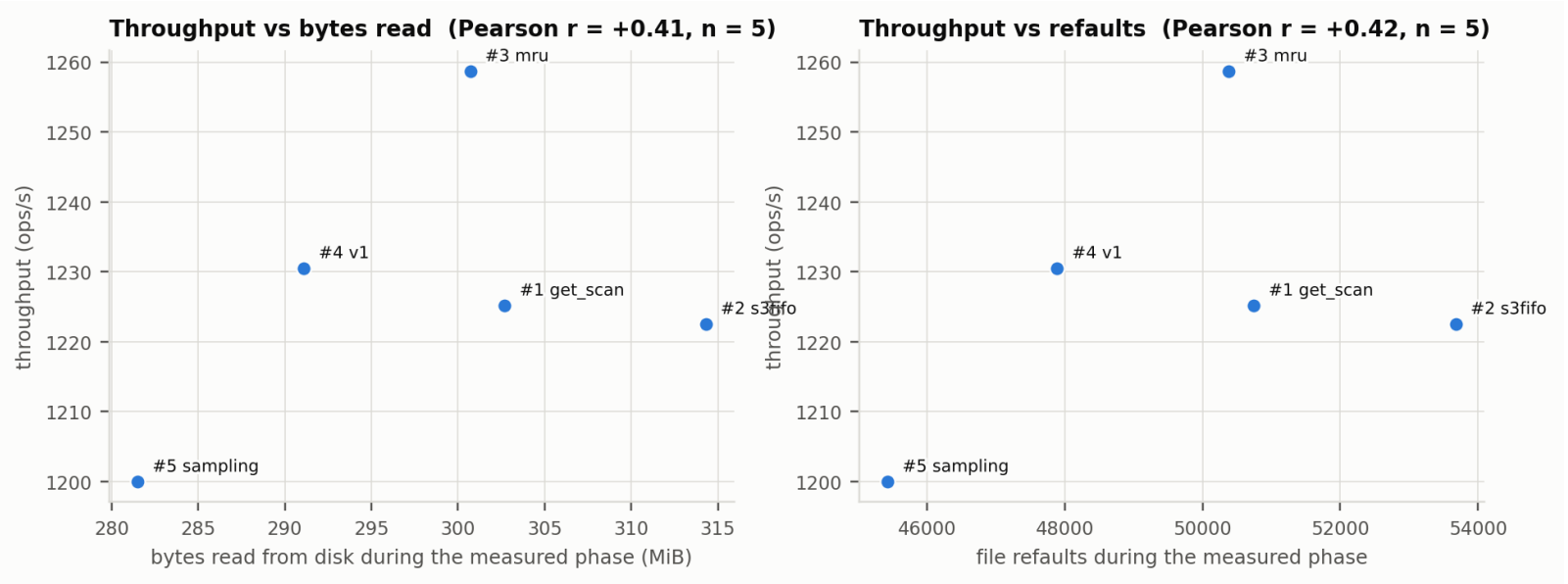
External run-1 noop reference for these probes: bytes read 304.1 MiB, refaults 51,054. Assumes the probes cover exactly the 30 s measured phase, so operations = throughput × 30.

Attempt descriptions (commit messages, verbatim)

#	Commit	Policy	Description
1	17a0a2bd	get_scan.c (unmodified)	Baseline: reference get_scan.c unchanged (scan_pids list + sampled LFU) to see if scan_pids is populated and how it compares to noop ~1215
2	938ee028	s3fifo.c (unmodified)	Baseline: reference s3fifo.c unchanged (small FIFO + main CLOCK + ghost), no pid info
3	b171f393	mru.c (unmodified)	Sensitivity probe: reference mru.c (should be bad for skewed GETs). get_scan=1225, s3fifo=1222, noop~1215 all identical; testing whether eviction policy moves throughput at all. (notes dir not writable from this agent: permission denied)
4	0e985cef	evolved v1	v1: scan_list (scan_pids OR per-thread sequential-add detector, score>=16) evicted first FIFO; GET-touched scan pages promoted; main LRU. MRU ref scored 1258 vs ~1222 for get_scan/s3fifo
5	1a7504b5	sampling.c (unmodified)	Datapoint: reference sampling.c (sampled LFU, 20 samples, index-block bonus). v1 scan-aware LRU got 1230 w/ lowest rbytes 291MiB; MRU 1258; testing whether frequency helps on Zipfian GET set

Score per attempt





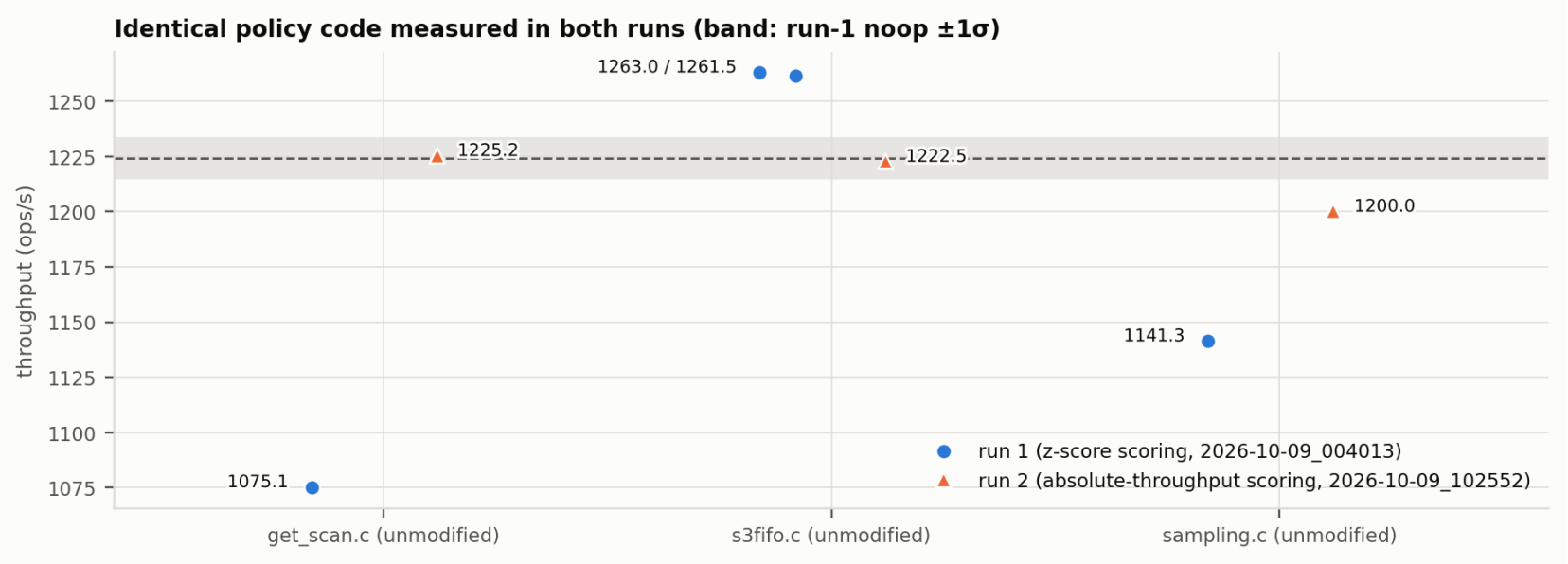
Same code, two runs

Policy (unmodified reference code)	Run 1 attempts (class: ops/s)	Run 1 mean	Run 2 (attempt: ops/s)	Δ ops/s (run 2 − run 1 mean)	Δ%	Δ in robust within-code sd	Run 1 bytes read MiB	Run 2 bytes read MiB	Run 1 refaults	Run 2 refaults
get_scan.c	real #1: 1,075.1	1,075.1	#1: 1,225.18	+150.0	+14.0	+22.9	257.7	302.7	39,626	50,745
s3fifo.c	real #2: 1,263.0, real #6: 1,261.5	1,262.2	#2: 1,222.54	-39.7	-3.1	-6.1	298.4	314.3	49,699	53,676
mglru.c	real #3: 1,257.6	1,257.6	not run	-	-	-	302.3	-	50,720	-
sampling.c	tune #19: 1,141.3	1,141.3	#5: 1,199.98	+58.7	+5.1	+8.9	270.5	281.5	42,743	45,427
lhd.c	tune #25: 1,070.5	1,070.5	not run	-	-	-	265.9	-	41,698	-
mru.c	not run	-	#3: 1,258.70	-	-	-	-	300.7	-	50,381

Run 1 = 2026-10-09_004013 (z-score scoring, 20-attempt cap); run 2 = this run. The attempts compared run byte-identical policy code (comment-only lines ignored): the agent copied each reference file over noop.c unchanged in both runs. mglru.c and lhd.c were run only in run 1, mru.c only in run 2.

Run-1 within-code spread (context for the size of the differences)

Quantity (from run 1: identical code measured more than once)	Value
Groups / measurements	3 groups, 8 measurements
Pooled within-code sd of throughput (all data)	34.85 ops/s (df 5)
Pooled within-code sd excluding the most deviant single measurement ('robust')	6.56 ops/s (df 4)
Largest within-code range in run 1	7.72 % of the group mean
Run-1 noop baseline sd (8 consecutive evals)	9.52 ops/s



Best attempt and best agent-evolved design

Item	Highest score (overall)	Best agent-evolved design
Attempt	#3 b171f393ceafdec2717e21179d65eb849cc3144a	#4 0e985cefaacc0e6146c4374747e6c497f8ac4480
Policy	mru.c (unmodified)	evolved v1 (agent's own design; no reference code)
Score (= throughput, ops/s)	1,258.70	1,230.52
vs external run-1 noop mean	+34.4 ops/s (+2.81%)	+6.2 ops/s (+0.51%)
Bytes read (MiB) / refaults	300.7 / 50,381	291.1 / 47,889
Code size	303 lines (kernel 132, loader 171)	369 lines (kernel 181, loader 188)
Description	Sensitivity probe: reference mru.c (should be bad for skewed GETs). get_scan=1225, s3fifo=1222, noop~1215 all identical; testing whether eviction policy moves throughput at all. (notes dir not writable from this agent: permission denied)	v1: scan_list (scan_pids OR per-thread sequential-add detector, score>=16) evicted first FIFO; GET-touched scan pages promoted; main LRU. MRU ref scored 1258 vs ~1222 for get_scan/s3fifo

Ranking of all attempts by score

Rank	#	Commit	Policy	Score (ops/s)	Δ vs best (ops/s)	Δ vs best (%)
1	3	b171f393	mru.c (unmodified)	1,258.70	+0.0	+0.00
2	4	0e985cef	evolved v1	1,230.52	-28.2	-2.24
3	1	17a0a2bd	get_scan.c (unmodified)	1,225.18	-33.5	-2.66
4	2	938ee028	s3fifo.c (unmodified)	1,222.54	-36.2	-2.87
5	5	1a7504b5	sampling.c (unmodified)	1,199.98	-58.7	-4.67

Each attempt is a single measurement; identical code varied by up to 7.7% in run 1, so adjacent ranks (differences under about 1–2%) are not distinguishable. The agent-evolved design's source is saved as best_evolved_policy.c.

Timing

Event	Time	Elapsed since start (min)
Run directory created / manager started	10:25:52	0.0
First model request at the gateway	10:25:58	0.1
First submission (attempt #1)	10:26:45	0.9
Last submission (attempt #5)	10:34:29	8.6
Last attempt graded / auto-stop recorded	10:35:17	9.4

Time budget	Seconds	% of run wall time
Evaluation (all 5 attempts, submit → graded)	228	40.3
Everything else (agent reading, thinking, editing)	338	59.7
Run wall time	**566**	**100.0**

Eval latency per attempt: min 43.9 s, median 46.0 s, max 46.1 s.

Agent sessions and LLM usage

#	Start	Wall s	API s (Δ)	Turns	Cost $ (Δ)	Input tok	Cache write tok	Cache read tok	Output tok	Thinking tok	API retries	Denials	Terminal reason
0	10:25:56	95	53	9	0.463	14	72,407	365,732	1,390	422	0	0	aborted_streaming
1	10:27:32	160	107	13	0.363	24	13,964	879,832	5,852	3,690	0	2	aborted_streaming
2	10:30:12	70	10	3	0.050	2	3,965	87,016	620	250	0	0	aborted_streaming
3	10:31:22	145	86	6	0.233	10	9,558	378,532	5,480	2,000	0	0	aborted_streaming
4	10:33:47	90	30	3	0.060	4	1,578	101,837	1,585	1,216	0	0	aborted_streaming
Σ		**559**	**285**	**34**	**1.17**	**54**	**101,472**	**1,812,949**	**14,927**	**7,578**	**0**	**2**	

Cost and API time are cumulative in Claude Code's result records, so rows show the increase over the previous session and Σ the final value; cost is Claude Code's estimate, not an AWS bill. Wall s, turns and tokens are per session. aborted_streaming = the session was cut off mid-response (consistent with CORAL interrupting and resuming the agent).

Agent actions

Tool	Calls
Bash	23
Write	1

coral command	Count
eval	6
log	3
checkout	2
wait	1
diff	1

Permission denials (actions the permission layer refused)

Session	Tool	Target (truncated)
1	Bash	{"command": "mkdir -p .claude/notes/experiments && python .claude/skills/create-notes/scripts/stamp.py experiment --out .claude/notes/experiments/eval-1-getscan
1	Write	{"file_path": "/home/aanil/Desktop/ldos/arjua-forks/coral-cache-evolution/results/cache-ext-get-scan/2026-10-09_102552/agents/captain-nemo/.claude/notes/experim

LLM gateway (LiteLLM → Bedrock)

Requests by model and HTTP status

Model alias	200	400	Total	Success %
claude-opus-5-5	40	5	45	88.9
claude-sonnet-5	0	5	5	0.0
all	**40**	**10**	**50**	**80.0**

Failure causes

HTTP	Model alias	Count	Message (summarised from the gateway's error body)
400	claude-opus-5-5	5	messages.N.output_config: extra inputs not permitted
400	claude-sonnet-5	5	thinking.type.disabled not supported for this model

Response time of successful (200) requests

n	Mean s	p50 s	p90 s	p99 s	Max s	Sum s
40	9.82	5.73	22.89	29.84	30.21	393

Context the agent received

Instruction file injected into the agent's prompts	Requests containing it	of total
~/arjua-forks/CLAUDE.md	50	50
~/arjua-forks/…/2026-10-09_102552/agents/captain-nemo/CLAUDE.md	50	50

Gateway: 50 requests in 6 gateway sessions. The first file is the arjua-forks repository's own project instructions (the earlier throughput-profiling project), loaded because the agent's worktree sits below that directory; the second is the task's CLAUDE.md written by CORAL.

Measurements not available for this run

Quantity	Status
Noop baseline in this run	None evaluated; the agent's own comparison point was the figure given in the task description (≈ 1210–1225 ops/s). The run-1 calibration is used as an external reference.
Per-operation latency (READ p99, SCAN p99), per-thread throughput	Written to a per-eval scratch directory that is deleted after each eval; not retained
Policy counters	No policy exported a metrics file
Agent notes	0 written; the agent's attempts to create the notes directory and files were denied (see permission denials)
CPU utilisation / cycles during evals	Not collected